



Cognify: Toy Design Competition

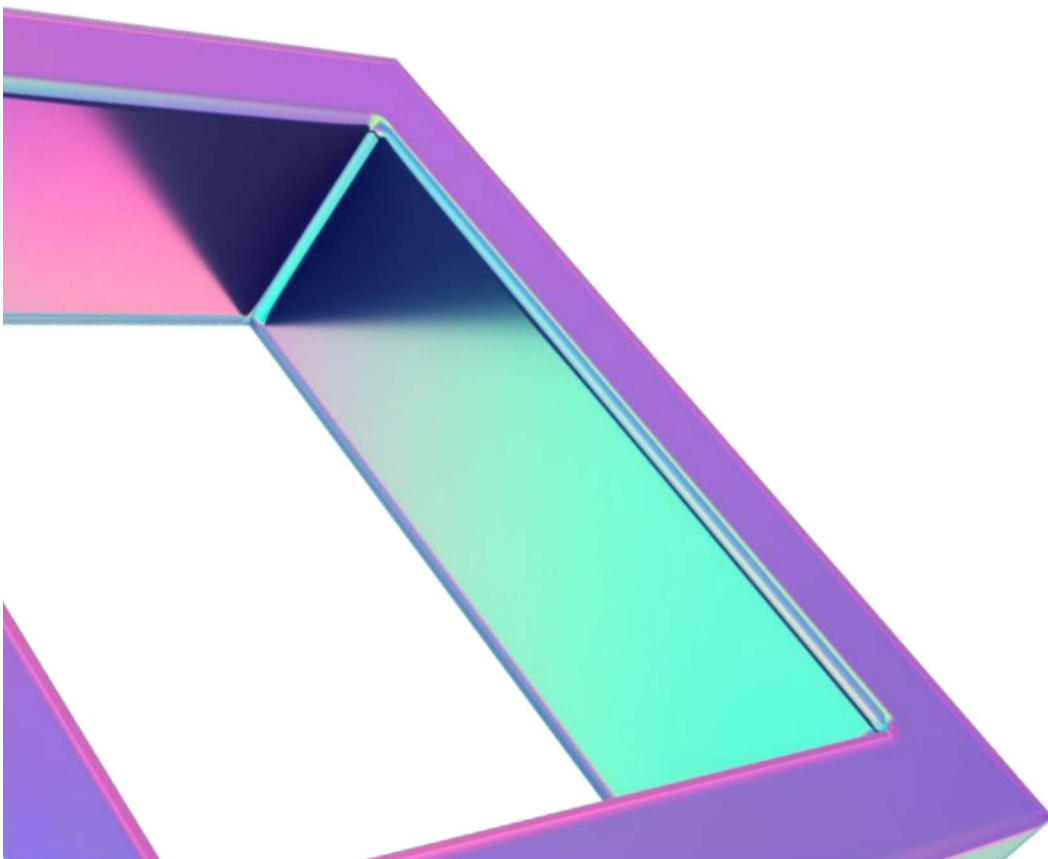
Round 2 : Problem statement

Team Amrita

Date: 6/12/2024

Table of Contents

Oscilipillar.....	2
3D and 2D Sketches.....	2
Structure and Design.....	5
Materials.....	8
Approximate Cost.....	8
Manufacturing Process.....	9
Sensory Responsiveness.....	9
Cognitive and Motor Skills.....	10
Playability.....	11
Acceptability.....	11
Parental and Educator Use.....	12
Novelty.....	13
Validation of Impact.....	13
References.....	14



Oscilipillar

3D and 2D Sketches

This toy is a modular, hollow-framed structure with colourful glitter-filled bobs that oscillate like pendulums. Swing Starter initiates the motion, while a lock mechanism secures the bobs after play. The top features a foam-covered lock for tactile interaction, and a sound element provides reinforcement with soothing tones, encouraging replay. It's designed to be engaging, visually captivating, and easy to interact with.

Target Users

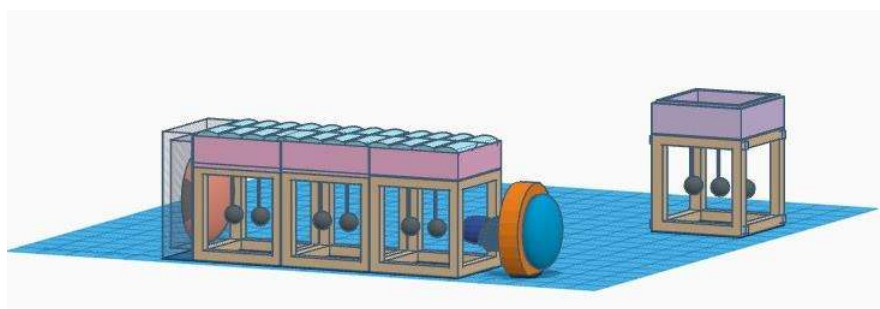
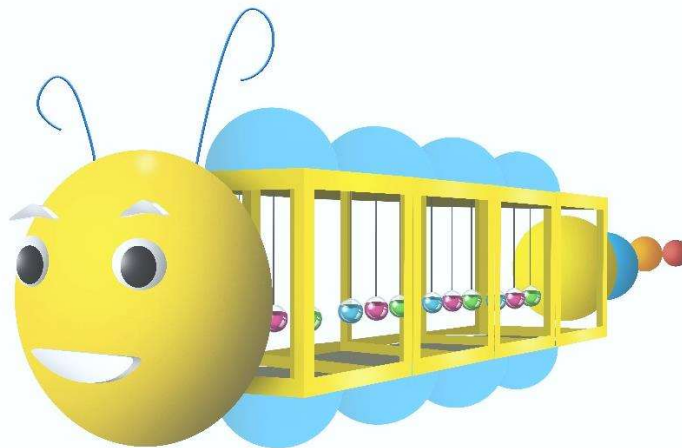
Age: 3-5

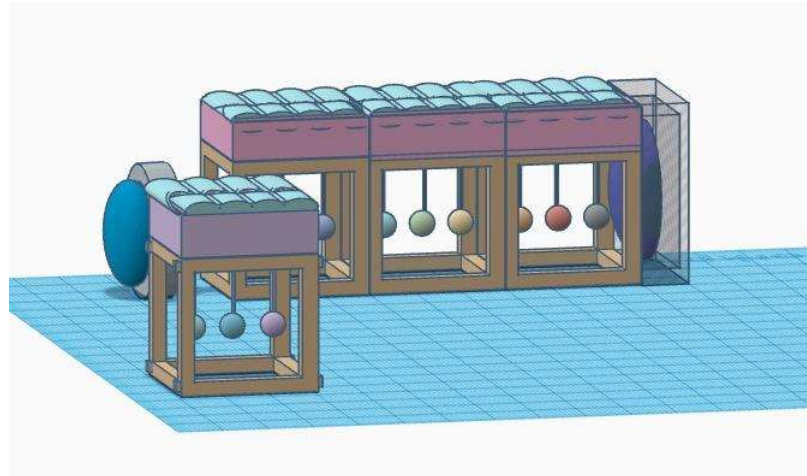
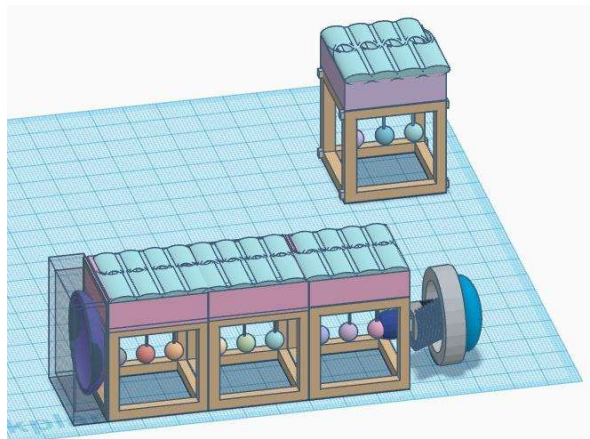
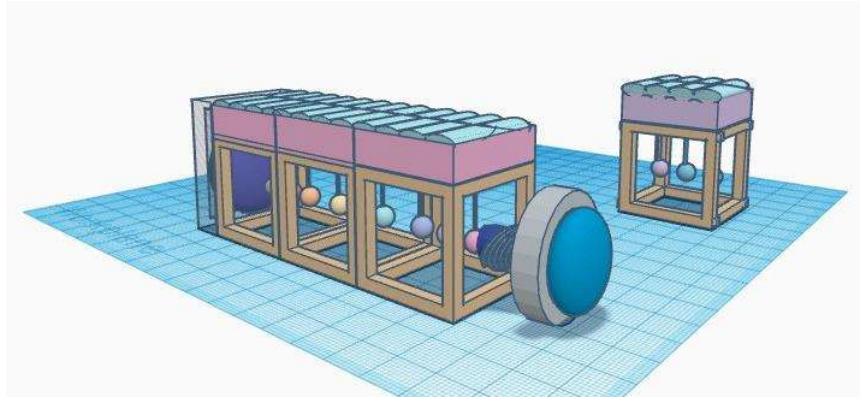
Could be used as they grow older by changing front piece of the toy to something which is of their interest (for e.g., train, crocodile).

Inclusive for all: for neurotypicals and neurodivergents

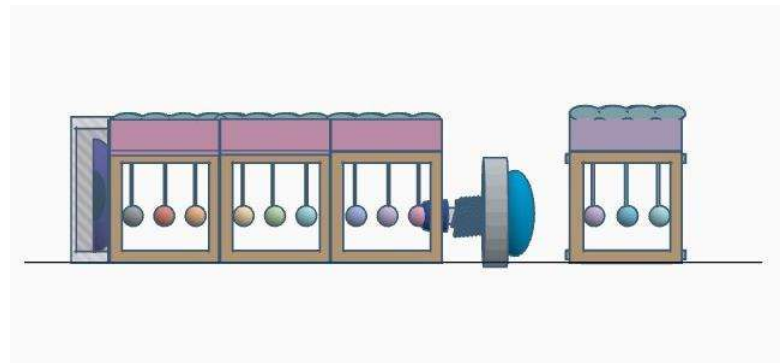
Purpose: Improving attentional faculties by incorporating coordinated saccades

3D Perspectives

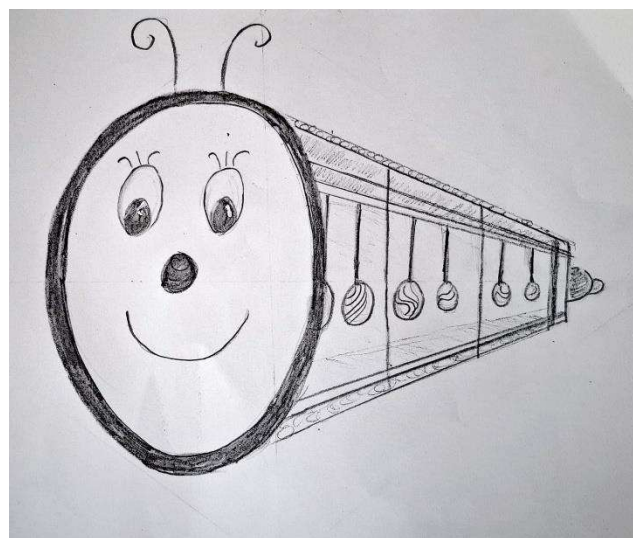




Side View

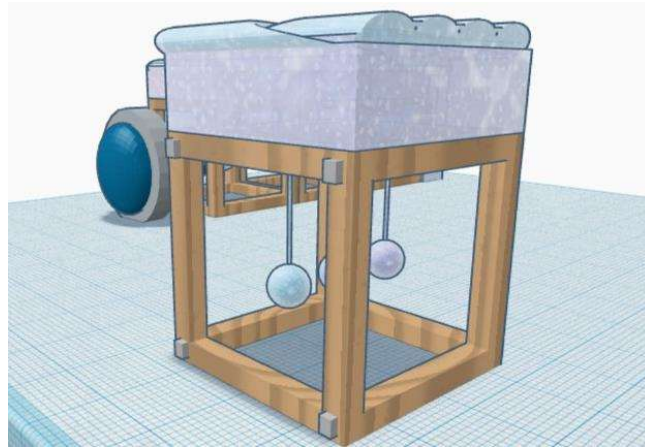


Sketch

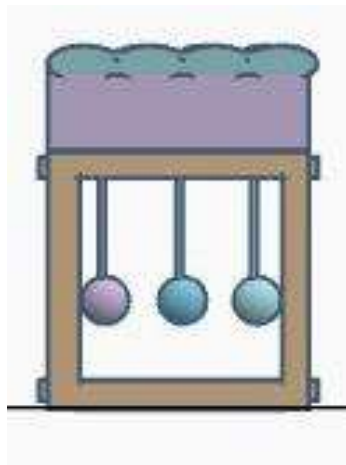
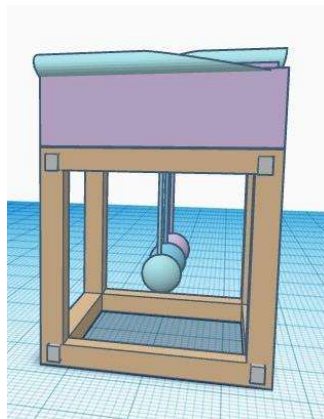


Structure and Design

Single Detachable Cube



The toy is a hollow-framed structure with detachable cubes. The cubes can be attached using magnets, allowing modular customization and easy assembly. The frame is see-through, enabling the player to easily observe and interact with the components during play.



Pendulum Mechanism

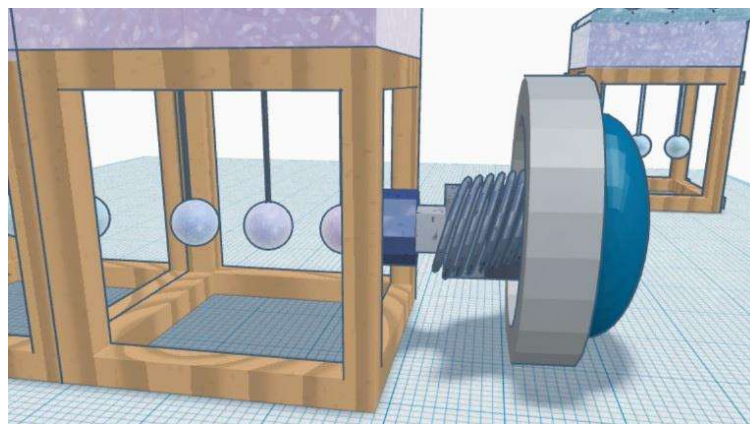
The toy features bobs that oscillate like pendulums, promoting smooth motion.

- The oscillation is designed to support **Saccadic eye movement**, adding a therapeutic and calming benefit.
(Merzon et al., 2022)
- Each bob is filled with **glittery jelly liquid** that enhances the visual appeal and creates a mesmerizing sense of motion.
- The **bobs come in different colours** and are suspended by strings for a seamless swing mechanism.



Swing Starter

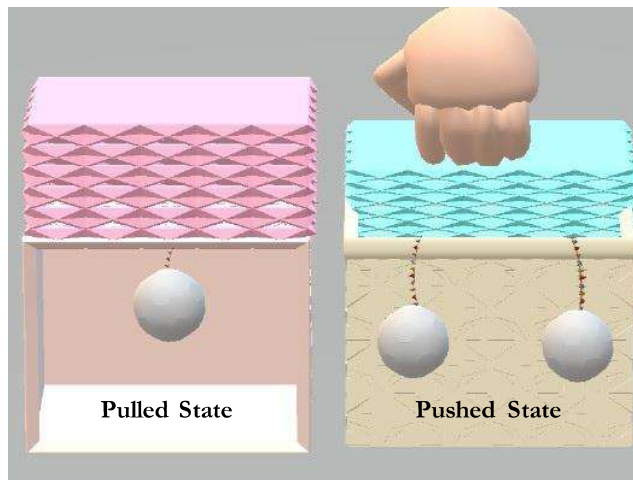
A button provides the initial push to set the first bob in motion, ensuring ease of use. The mechanism includes a spring system that allows the button to return to its initial position automatically after being pushed. The button's outer casing is customizable with fun designs to engage the player further.



Locking and Protective Mechanisms

- The top section of the toy features a crisscross lock mechanism.
- **Pulled State:** Locks the bobs in place for secure storage after play.
- **Pushed State:** Suspends the bobs to protect them from damage during movement.

The locking mechanism is covered with a foam sheet, providing tactile sensory feedback and comfort to the player.

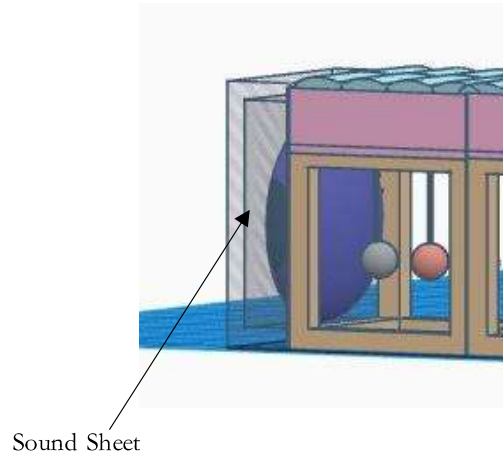


Auditory Stimuli

The toy includes an additional element with a music or sound-producing sheet. This sheet creates a soothing, tinkling sound effect, providing reinforcement and feedback to the player. The sound effect is designed to signal the end of a round when all the bobs have struck each other, encouraging the player to replay and enjoy the experience.

This is detachable with two options: Wind chime and drum. This gives the children an opportunity to explore and experiment with the sounds of both the sheet.

(Shahrudin et al., 2022)



Materials

The material which would be used to make the toy are:

- Acrylic sheet – Transparent cubes
- Good grade plastic- Lid and criss-cross screws
- Foam- Body of the toy
- Dymeena- String
- Rubber- Hammer
- Magnet- Within the Edges of the cube

Approximate Cost

Cost of the materials and total cost (excluding labour charge):

- Acrylic sheet - Rs.600 (Surface area-150 square cm)
- Good grade plastic- Rs.250 (Surface area- 100 square cm)
- Foam- Body of the toy-Rs.100
- Dymeena- String- Rs.80(40cm)
- Rubber- Hammer -Rs. 20
- Magnet- Within the Edges of the cube-Rs. 200(24 small magnets)
- Approximate Total cost= Rs. 1250

Manufacturing Process

Making of the Cubes:

- **Injection Molding:** acrylic granules are melted and injected into a mold to form the toy parts.
- **Assembly:** The molded pieces are trimmed, painted, and assembled.
- **Finishing:** Smooth edges, apply paints or decals, and add fine details.

Making of the string and bob inside the cube:

- **Prepare the Dyneema Rope:**
 - 1) Cut the rope to the required length and seal the ends with heat (use a lighter or flame). This prevents fraying.
 - 2) Tie a figure-eight knot or an overhand loop at the end of the rope to prevent the bob from sliding off.
- **Thread the Rope Through the Bob:** Pass the rope through the hole in the marble bob. Slide the bob until it rests against the knot.
- **Secure the Knot :** Apply a small amount of superglue or epoxy resin to the knot to ensure it doesn't slip under tension.

Sensory Responsiveness

- Foam used in the head, tail and body of the toy gives the tactile sensory comfort.
- The Glittery coloured liquid is visually appealing and enhances the visual sensory engagement towards the pendular oscillatory pattern.
- Auditory appeal is achieved by attaching different sheets to the inner surface of the tail. For example: A thin sheet of mango wood or Metal used in wind chimes. The bob strikes the Sheet and produces a soothing sound.

Cognitive and Motor Skills

The child starts attaching the cubes starting from head and ending at the tail. With a couple of tries, the child learns to align the cubes in a specific order to form the whole structure. This Activity enhances the gross motor skills.

To start the oscillation, the child clicks the button at the head of the toy. This enhances the fine motor skill. Additionally, the major motive of the toy is to develop well-coordinated saccades. This is a blend of cognitive(Attention) and motor(coordinated eye movement) skills.

Theories from Psychology

- **Attention: Selective and Cued Attention**

The toy aims to utilize strengths of neurodivergent individuals to exercise saccadic eye movement. According to the theory of monotropism for autism, the autistic person is theorized to see the world through a narrow attentional tunnel. This theory may also be used to explain hyperfocus in ADHD individuals. (Dwyer et al., 2024). Specifically for autistic individuals, monotropism theory may imply that autistic individuals prefer a more 'local perspective than 'global' perspective (Amudha et al., 2020; Murray et al., 2005). Through this toy, we would utilize the strength of monotropism as the children who engage with the toy would be observing the pendular movement of the balls hitting against each other. The movement of the balls, the movement of the liquid, the sound made by the balls would help in cueing the children's attention towards the ball.

- **Sensory Memory**

Sensory memory (visual and sound) is an inherent cognitive function which would take place when the children are observing the flow of the balls. This pattern may also help in drawing attention.

- **Reinforcement (Operant Conditioning)**

As the child is exploring the toy, the child may assume positive reinforcement when the boxes fit through magnets and when there is a movement through a push of the button. In addition, the sounds at the end may also be considered reward as they would ring when the balls of the pendulum hit it. To get this reward, the child may do the action of pushing of the button again. Hence, this may help in increasing the child's engagement with toy. This phenomena is hypothesized in reference to Skinner's theory of operant conditioning (Skinner, 1963).

- **Delayed Gratification**

If we could at these cognitive functions in the perspective of figure ground in gestalt psychology, then attention would be in the foreground and reinforcement and delayed gratification would be in the background. The theory of delayed gratification was developed by Mischel and colleagues through a Marshmallow experiment in children (Mischel et al., 1989). In this theory, delayed gratification refers to when immediate reward is postponed to achieve a later reward. In respect to the toy, it could be hypothesized that children would have to wait till the last ball hits the sound sheet to get the full effect. Even if it just a matter of second, waiting till the last ball hits could help in building impulse control and self-regulation.

Playability

The toy offers a well-rounded play experience with a balance of visual stimulation, interactive mechanics, and sensory feedback. The combination of simple motion and calming features makes it ideal for relaxation and focus, while the sound and modular design encourage prolonged engagement and repeated play.

Visual Appeal

- The transparent design and colourful glitter-filled bobs will quickly attract the player's attention, creating a visually interesting experience. Players can watch the bobs move back and forth, which can engage them both intellectually and visually.

Sequential and Interactive Play:

- **Button Mechanism:** The starting push button allows players to easily initiate the oscillation, giving them a sense of control. This can be satisfying, as the player sees the chain reaction of bobs hitting each other in sequence.
- **Sound Reinforcement:** The tinkling sound effect will encourage players to wait for the sequence to complete, giving them a sense of progression and reward when the round ends. This also reinforces their engagement, encouraging them to play again.
- **Exploratory Play:** Can explore by experimenting through different sound sheets, different cubes and joining them together

Acceptability

The Toy complies to the narratives of neurodivergence movement by utilizing **monotropism** as a strength. It does not focus on transforming a neurodivergent child to a neurotypical child. The Toy rather acts a medium to execute and exhibit their strength of being **hyperfocused on the specific details of an object** not **the global structure**. Thus, the toy is **user-centric**.

Parental and Educator Use

Child's Safety

It ensures child's safety through

- Protective gear and lock mechanism.
- Strong materials used for string so that it does not break apart.

Sustainable material:

Materials selected to make the toy would be eco-friendly/recyclable and child-safe. The cubes could be reused as it could be used with different front-parts/back-parts suiting their interests as they grow up (for e.g. train, crocodile)

Therapeutic Aspect

Multi-modal engagement, integrating all essential cognitive skills as part of early development. It may help in practicing the cognitive function of attention during periods of neuroplasticity.

These eye exercises may assist in improving saccades.

Educational Aspect

The toy can enhance embodied learning and improves spatial cognition (Mental rotation and Physical rotation) to form the whole structure.

Educators could use it to show movements such as

- Up-down
- Left-right
- Push-pull
- Cause-effect

Novelty

- **Combination of Glitter-Filled Bobs & Pendulum Motion:** While there are pendulum toys, the use of glitter jelly liquid to enhance the visual motion is a distinct feature.
- **Tactile, Modular & Magnetic Design:** The ability to detach and reassemble cubes via magnets adds a customizable, hands-on element that isn't common in most traditional toys.
- **Sound Reinforcement for Sequence Completion:** The combination of soothing sounds with interactive play (waiting for bobs to hit in sequence) is unique in that it provides both auditory and visual feedback to reinforce play.
- **Therapeutic Features:** The integration of saccadic eye movement support adds a deeper, more beneficial component to the toy, making it stand out from typical entertainment-focused products.
- **Customizable:** Later, could be customized using different front-parts/back-parts which may of interest to the player.

Validation of Impact

- The target age group is 3-5 years. This is the crucial developmental stage where the child exhibits the pattern of neurodivergence.
- The toy can act as an early developmental tool to enhance structural plasticity.
- Sensory -Motor Integration therapy is a commonly used modality for all the neurodivergent children. The toy can be used as a therapeutic tool, is inclusive and can be modified according to the need of a child.

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